Richard Djarbeng

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Personal Statement

I am a research-oriented engineer with an MSc in Electrical and Computer Engineering from Carnegie Mellon University. My academic and research journey has built a strong foundation in deep learning, computer vision, wireless networks, and their applications in autonomous systems and environmental sensing. My research roles have heavily focused on automation for Internet of Things (IoT), air quality analysis, and climate monitoring. Proficient in Python, C++, and machine learning frameworks (PyTorch, fastai), with practical experience developing robust data pipelines and deploying models on Google Cloud and Hugging Face. Through my tenure as a research associate and teaching assistant, I have developed strong capabilities in applied machine learning, data analysis, and hardware-software co-design to tackle real-world environmental challenges.

Education

- **Carnegie Mellon University** — July 2022 - May 2024
  MSc Electrical and Computer Engineering — *Pittsburgh, PA / Kigali, Rwanda*
  Courses: *Introduction to Deep Learning, Computer Vision, Internet of Things, Autonomous Driving, Data Analytics, Data Structures and Algorithms, Introduction to Information Security, Cognitive Robotics, Wireless Networks*

- **University of Ghana** — November 2021
  BSc Computer Engineering — *Accra, Ghana*
  Relevant Coursework: Software Engineering, Computer Graphics, Object Oriented Programming, C/C++ Programming, Robotics and Computer Vision, Data Structures and Algorithms, Introduction to Artificial Intelligence

Certifications & Professional Development

- **AWS Certified Solutions Architect - Associate** — June 2026
  Amazon Web Services
  Demonstrated ability to build secure, high-performing, and resilient cloud infrastructure for large-scale data and research applications.
- **Google Get Certified Program - Professional ML Engineer Track** — June 2024 - August 2024
  Google Cloud
  Gained applied experience in production ML workflows, including model training, monitoring, and MLOps using Vertex AI and BigQuery ML.

Research & Professional Experience

- **Research Associate** — July 2024 - October 2025
  Carnegie Mellon University — *Kigali, Rwanda*
  - **Environmental Sensor Development:** Designed and developed low-cost sensor nodes (PCBs, Fusion360 CAD) for measuring temperature, humidity, wind speed, and environmental parameters using innovative AI-driven calibration approaches.
  - **Sensor Testing and Calibration:** Applied machine learning techniques to improve the accuracy and reliability of low-cost sensors, benchmarking against high-grade reference instruments.
  - **Data Storage and Management:** Collaborated with principal investigators and external researchers to design a secure data storage and management framework, ensuring collected environmental data was accessible to project stakeholders.
  - **Air Quality Data Analysis:** Analyzed air quality data from US Embassy sources in Kigali and other cities via Airnow, creating visualizations leveraging archived datasets for enhanced environmental insights.
  - **Academic Publication:** Collaborated with the research team to prepare academic papers documenting the project's findings and methodologies.
- **Pre-doctoral Research School in Computer Science (CMMRS)** — August 2025
  Cornell, Maryland, Max Planck — *Saarbrücken, Germany*

- **Pre-doctoral Development:** Selected for a merit-based, intensive research program designed to demystify the doctoral journey and foster advanced research pursuits.
- **State-of-the-Art Research Analysis:** Engaged in specialized lectures on computer vision, high-performance computing, and computer architecture, synthesizing intersecting research outcomes across various academic groups.
- **Academic & Career Strategy:** Participated in panels with researchers from Cornell, UMD, and Max Planck to evaluate transcontinental PhD structures and industry versus academia pathways.

- **Research Assistant** — January 2024 - May 2024
  Carnegie Mellon University — *Kigali, Rwanda*
  - **Interdisciplinary Research:** Worked with the mechanical engineering department and Center for Atmospheric Particle Studies (CAPS) to research alternative methods of predicting biomass pollutants using aethalometer sensor data.
  - **Pollution Source Classification:** Developed computer vision models to classify pollution sources (wood burning vs. fossil fuels) from RGB images of particulate matter filters captured via smartphone. Research featured on the CMU Africa website.
  - **Data Analysis and Experimentation:** Conducted comprehensive data analysis on datasets collected over more than 2 years and verified experiments to measure black carbon pollutants.
- **Teaching Assistant, Internet of Things** — August 2023 - December 2023
  Carnegie Mellon University — *Kigali, Rwanda*
  - **Teaching and Mentoring:** Conducted recitations focused on building IoT systems with web APIs using JavaScript and Node.js on the Raspberry Pi.
  - **Academic Support:** Assisted students in understanding circuit design, embedded systems programming, REST, HTTP, and wireless connectivity (2G, 3G).
- **Graduate Intern** — May 2023 - August 2023
  AfriqAir - Kigali Collaborative Research Center — *Kigali, Rwanda*
  - **System Design:** Designed system specifications and architecture for an IoT wireless data logging unit, coordinating requirements across hardware and software teams.
  - **Prototype Development:** Built a prototype IoT system to automate the diagnostic and data collection process from the Thermofisher TEOM™ air quality system using Python and C++ for Raspberry Pi and Arduino.
  - **Web Dashboard Development:** Developed a production-grade web dashboard using React.js and TypeScript for real-time visualization and interpretation of air quality data, consuming live sensor feeds.
- **Intern** — July 2020 - January 2021
  Sorphise GH Ltd — *Ghana*
  - **CMS Integration and Architecture:** Led the adoption of a headless CMS architecture decoupling content from code using NetlifyCMS, integrated with GitHub for CI/CD.
  - **Web Application Development:** Developed and deployed scalable web applications using JavaScript, PHP, React (GatsbyJS), and Netlify.

Technical Skills

- **Machine Learning & CV:** PyTorch, fastai, Scikit-learn, CNN architectures, ResNet, Semantic Segmentation, Object Detection, YOLO, Deepfake Detection, Model Deployment
- **Cloud & MLOps:** Google Cloud Platform (Vertex AI, AutoML, BigQuery ML), AWS, CI/CD, Hugging Face Spaces
- **Data & Analytics:** Python (Pandas, NumPy), SQL, BigQuery, Air Quality Data Analysis, Geospatial Analysis
- **Embedded & IoT:** C, C++, Raspberry Pi, Arduino, Sensor Calibration, GSM Modules, Wireless Connectivity
- **Web Development:** React.js, Node.js, TypeScript, JavaScript (ES6+), RESTful APIs, WebSockets
- **Languages:** Python, JavaScript, Java, C++, SQL, Dart (Flutter)

Leadership Experience

- **Club President, IoT Club** Carnegie Mellon University Africa (2022 - 2024)
  - Led a team of 6 committee members to organize campus events enhancing student knowledge of IoT and embedded systems, fostering a culture of innovation and technological exploration.

  - Orchestrated field trips to Zipline, a company pioneering autonomous drone delivery of medical supplies to remote areas of Rwanda, providing real-world exposure to cutting-edge AI and robotics in humanitarian contexts.

  - Served as a technical advisor for student projects, offering expertise in data collection, GPS integration, and microcontroller programming.

Selected Projects

- **Air Quality Analysis Dashboard** May 2024
  React.js, TypeScript, Data Visualization, Air Quality Modeling
  - **Trend Analysis:** Developed an interactive dashboard to analyze PM2.5 trends in Kigali, leveraging US Embassy reference data and planetary boundary layer dynamics to identify diurnal pollution patterns.
  - **Health Impact Quantification:** Quantified public health risks by identifying that 57% of readings exceeded WHO "Unhealthy" thresholds, with an average concentration of $43.31\,\mu g/m^3$.
  - **Validation:** Collaborated with academic advisors to validate model-observation evaluations and emission source meteorology.
- **Background Removal and Image Segmentation** August 2024
  Python, Deep Learning, Computer Vision, Gradio, Hugging Face
  - **Model Deployment:** Deployed a semantic segmentation model for HD background removal leveraging the `rembg` library into a production-ready Gradio interface on Hugging Face Spaces.
  - **Architecture Support:** Supported multiple model architectures (U-Net, ISNet, Silueta) to cover different segmentation workflows with specialized mask post-processing capabilities.
- **Predicting Loan Risk with AutoML** May 2024
  Google Cloud Vertex AI, AutoML, BigQuery, Explainable AI
  - **Predictive Modeling:** Developed an ML model to predict loan default risk using GCP Vertex AI's AutoML.
  - **Optimization & Evaluation:** Engineered features from tabular data, optimized training with early stopping, and evaluated using precision/recall curves and Explainable AI for feature importance.
- **Detection of Manipulated Facial Images** September 2023 - December 2023
  Python, Deep Learning, Computer Vision, ResNet, CNN
  - **Deepfake Detection:** Developed a robust system using deep learning techniques to detect and identify manipulated facial images, focusing on tools like Face Swap and Face2Face.
  - **Classification Accuracy:** Utilized a Block-based CNN and a ResNet-34 model, applying transfer learning to achieve high accuracy in classifying manipulated images extracted from videos.
- **Transmission Corridor Monitoring using Satellite Imagery** September 2020 - August 2021
  Python, Google Earth Engine, CNN, Satellite Image Processing
  - **Geospatial Analysis:** Developed a web application system for GRIDCo/ECG in Ghana to monitor power transmission corridors using satellite imagery.
  - **Hazard Detection:** Applied image processing and geospatial analysis using Google Earth Engine and CNNs to detect and alert on potential hazards or encroachments.